

The Internet

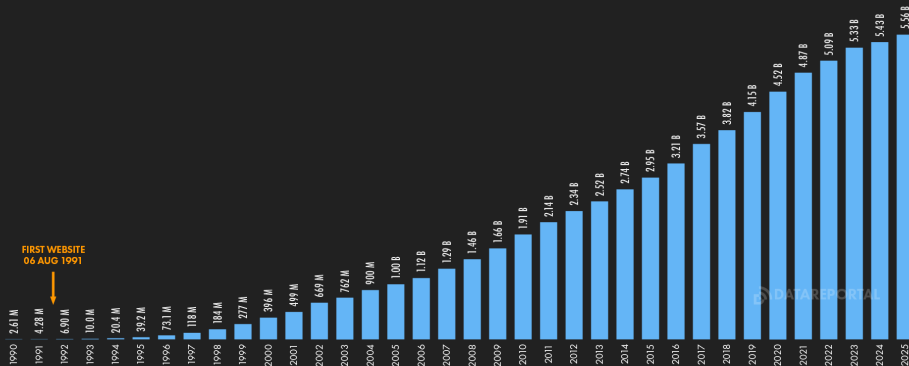
Perhaps the most successful computer application ever

Can you name a computer program that doesn't use the internet?

FEB
2025

INTERNET USE TIMELINE

NUMBER OF INDIVIDUALS USING THE INTERNET OVER TIME



48

SOURCES: KEPIOS ANALYSIS, ITU, GSMA INTELLIGENCE, EUROSTAT, GOOGLE'S ADVERTISING RESOURCES, CNNIC, KANTAR & IAMA, GOVERNMENT RESOURCES, UNITED NATIONS. COMPARABILITY: SOURCE AND BASE CHANGES. ALL FIGURES USE THE LATEST AVAILABLE DATA, BUT SOME SOURCES DO NOT PUBLISH REGULAR UPDATES, SO FIGURES FOR RECENT PERIODS MAY UNDER-REPRESENT ACTUAL USE. SEE NOTES ON DATA.

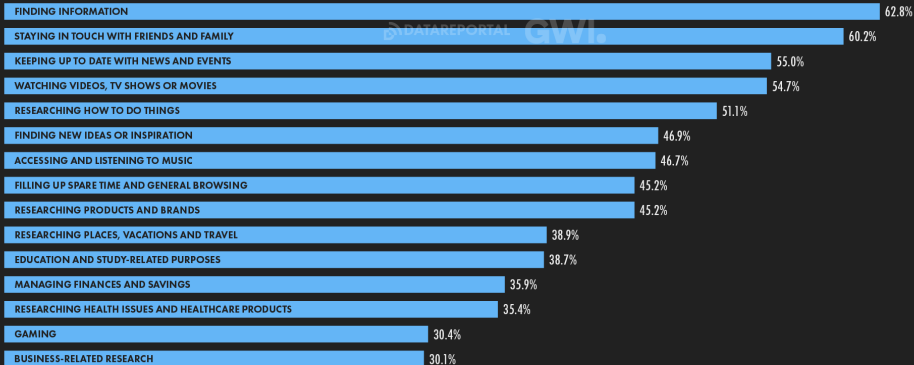
we
are
social

Meltwater

FEB
2025

MAIN REASONS FOR USING THE INTERNET

PRIMARY REASONS WHY INTERNET USERS AGED 16+ USE THE INTERNET



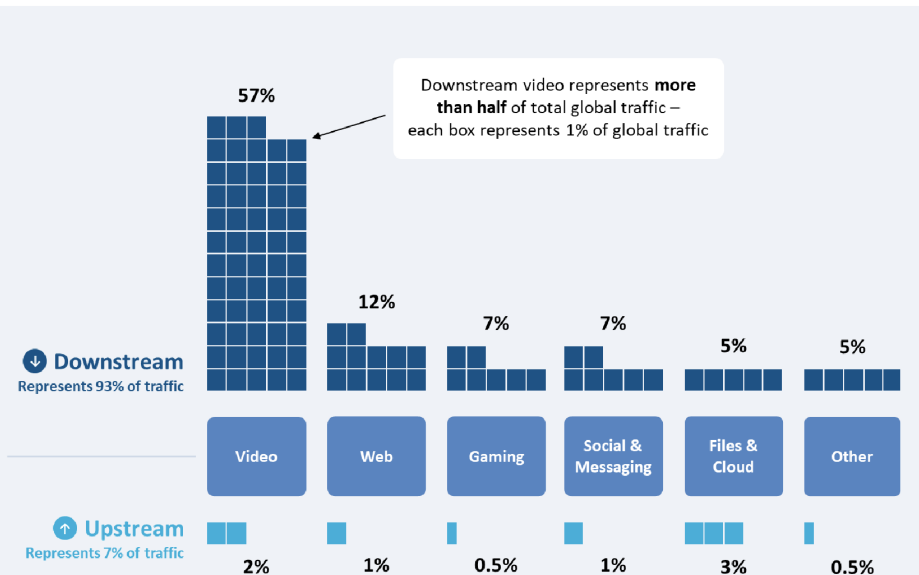
65

SOURCE: GWI (Q3 2024). COMPARABILITY: CHANGES IN AUDIENCE COMPOSITION AND SURVEY METHODOLOGY. SEE NOTES ON DATA.

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FIGURE 4. SHARE OF GLOBAL BANDWIDTH CONSUMED BY DIFFERENT INTERNET TRAFFIC TYPES



sockets is the de-facto API for networking

open connection then ...

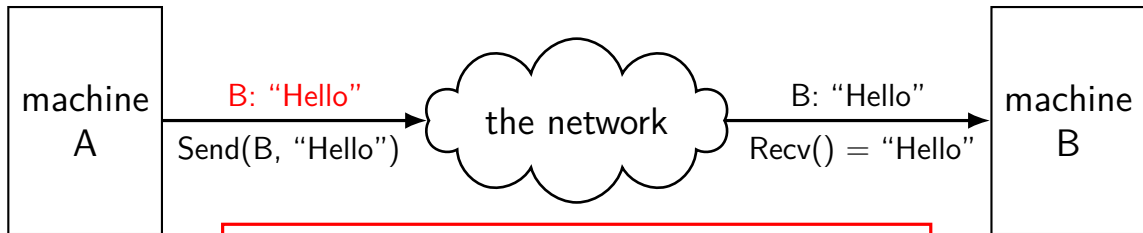
read+write just like a terminal file

doesn't look like individual messages

“connection abstraction”

mailbox model

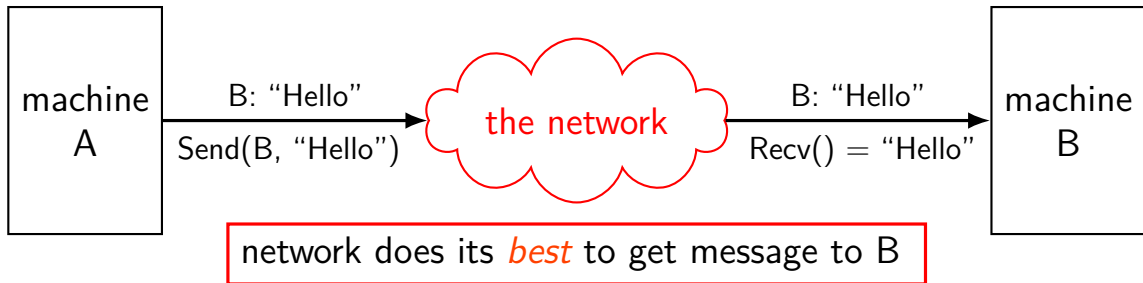
mailbox abstraction: send/receive messages



A sends "letter" to B
"envelope" tells network it's addressed to B
data in this example: "Hello"

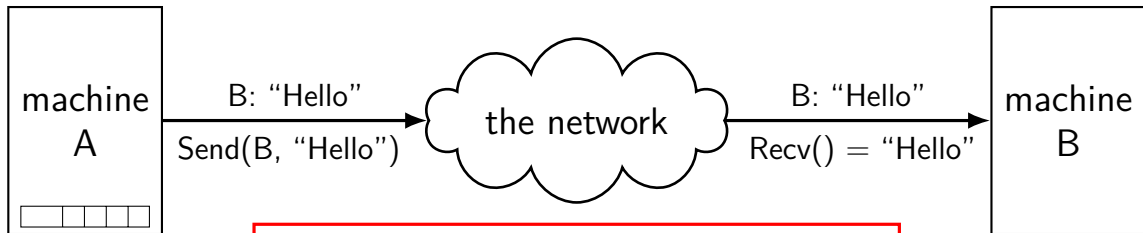
mailbox model

mailbox abstraction: send/receive messages



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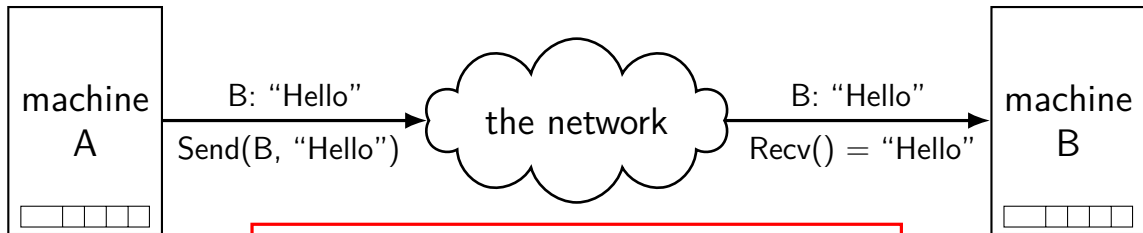
mailbox abstraction: send/receive messages



queue ('outgoing mailbox') of messages
from sending program
waiting to be sent

mailbox model

mailbox abstraction: send/receive messages



queue ('incoming mailbox') of messages
not yet received by
receiving program

connections over mailboxes

real Internet: mailbox-style communication

- send “letters” (packets) to particular mailboxes

- have “envelope” (header) saying where they go

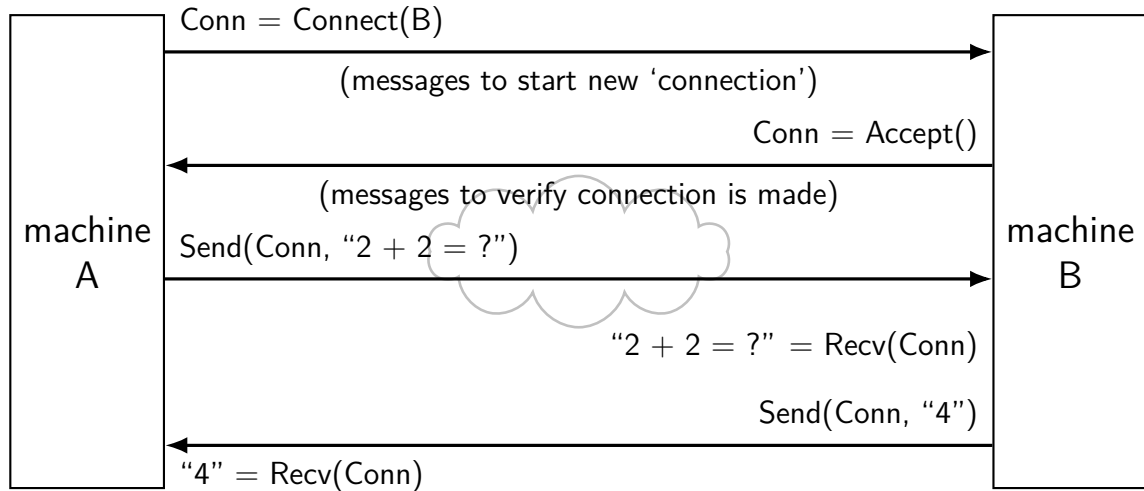
“best-effort”

- no guarantee on order, when received

- no guarantee on *if* received

sockets implemented on top of this

connections



networking layers

application	HTTP, SSH, SMTP, ...	application-defined meanings
transport	TCP, UDP, ...	reach correct program, reliability/streams
network	IPv4, IPv6	reach correct machine (across networks)
link	Ethernet, Wi-Fi, ...	coordinate shared wire/radio
physical	Ethernet, Wi-Fi, ...	encode bits for wire/radio

layers terminology

application	application-defined meanings	
transport	reach correct program, reliability/streams	segments/datagrams
network	reach correct machine (across networks)	packets
link	coordinate shared wire/radio	frames
physical	encode bits for wire/radio	

layer encapsulation

upper layers implemented using lower layers

example: implement reliable + large messages (transport layer)
by sending multiple unreliable messages across networks (network layer)

example: implement reaching machine across networks (network layer)
by sending multiple messages on local networks (link layer)

networking layers

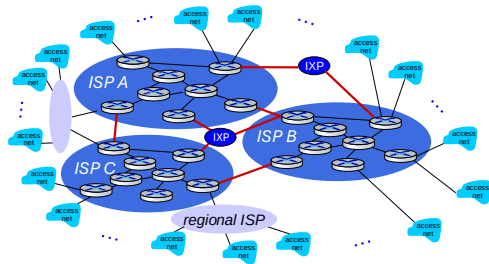
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reality: you don't own the network!

the internet is a series of connected networks

packets traverse routers within and between networks

those networks provide “best-effort” service



network limitations/failures

messages lost

messages delayed/reordered

messages limited in size

messages corrupted

network limitations/failures

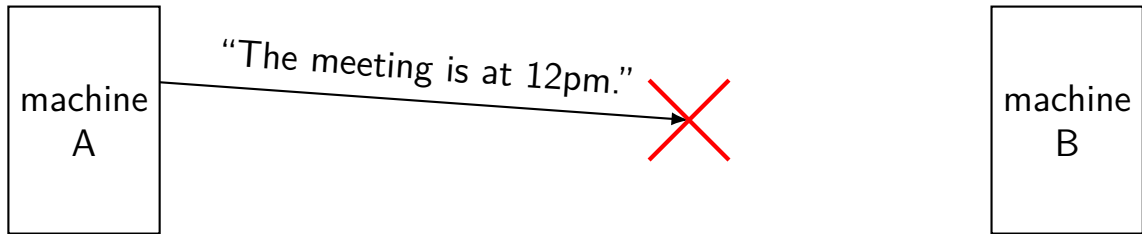
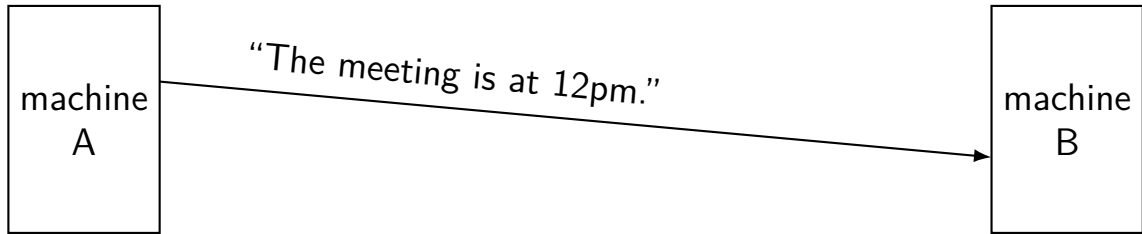
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messages delayed/reordered

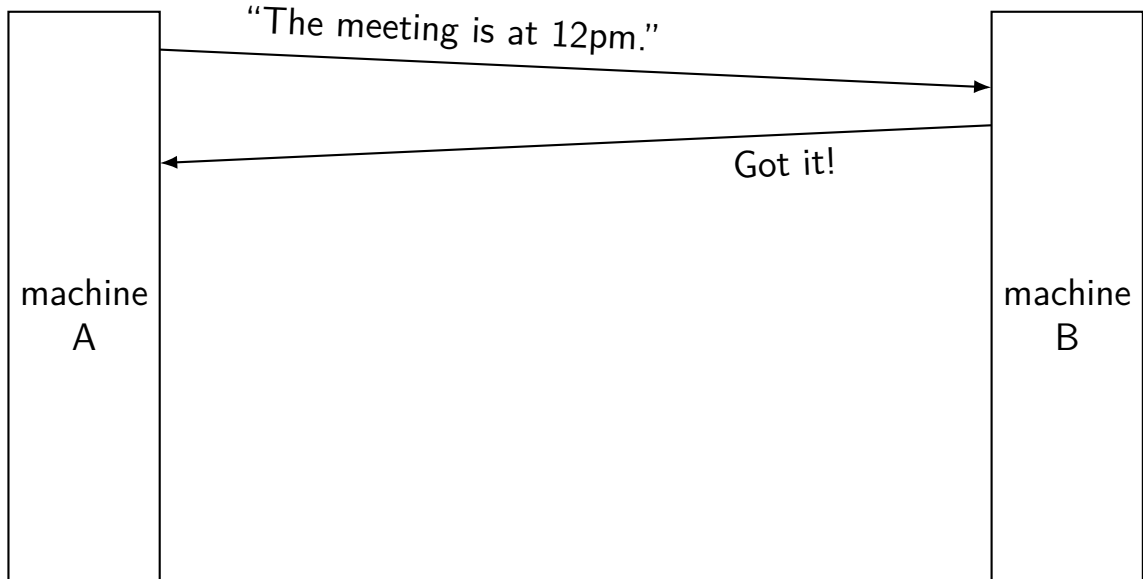
messages limited in size

messages corrupted

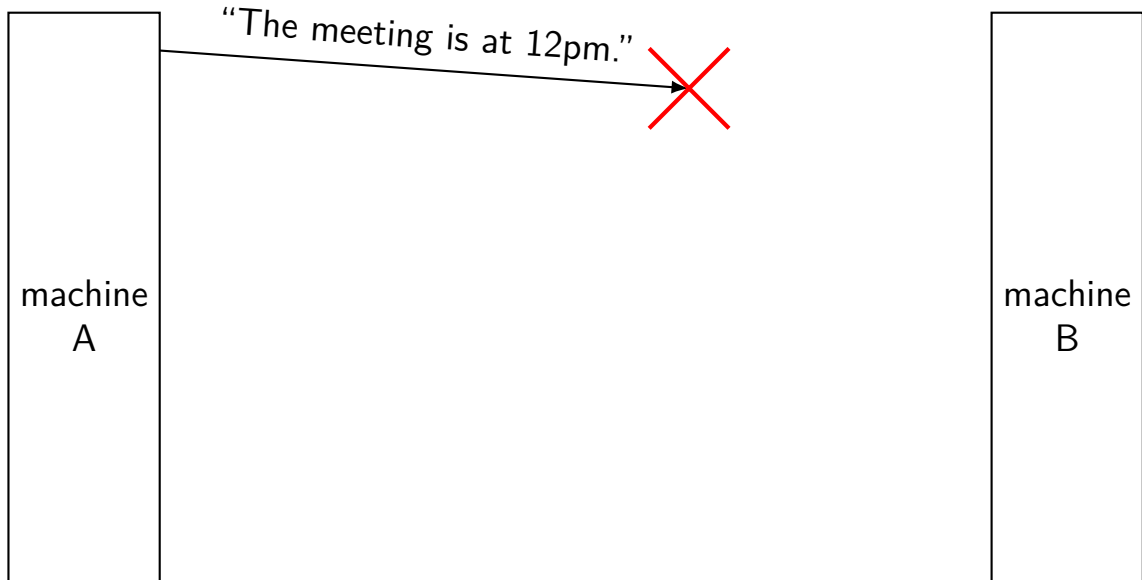
dealing with network message lost



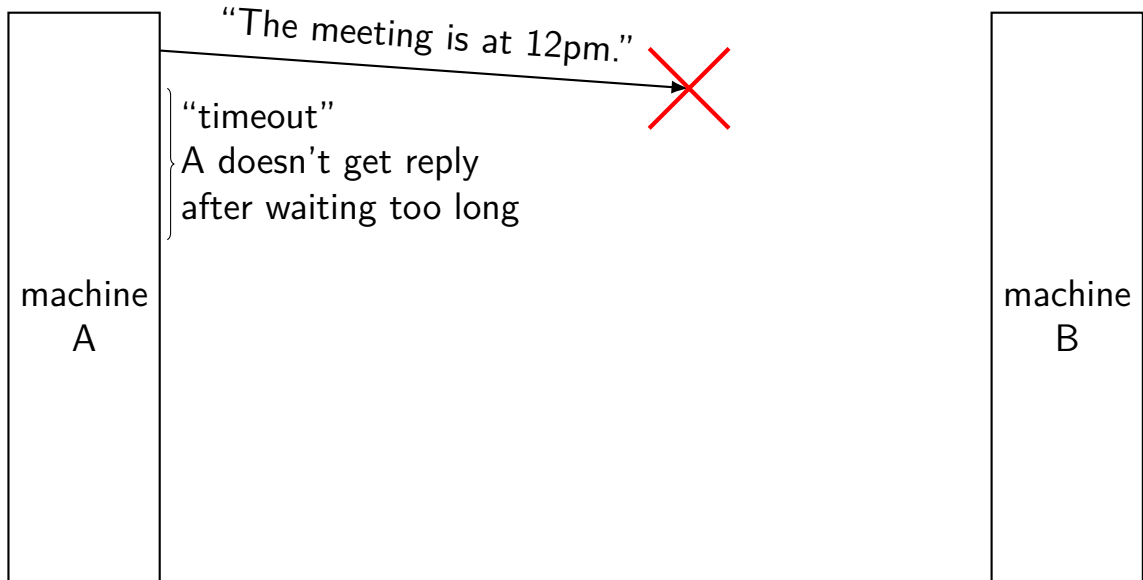
handling lost message: acknowledgements



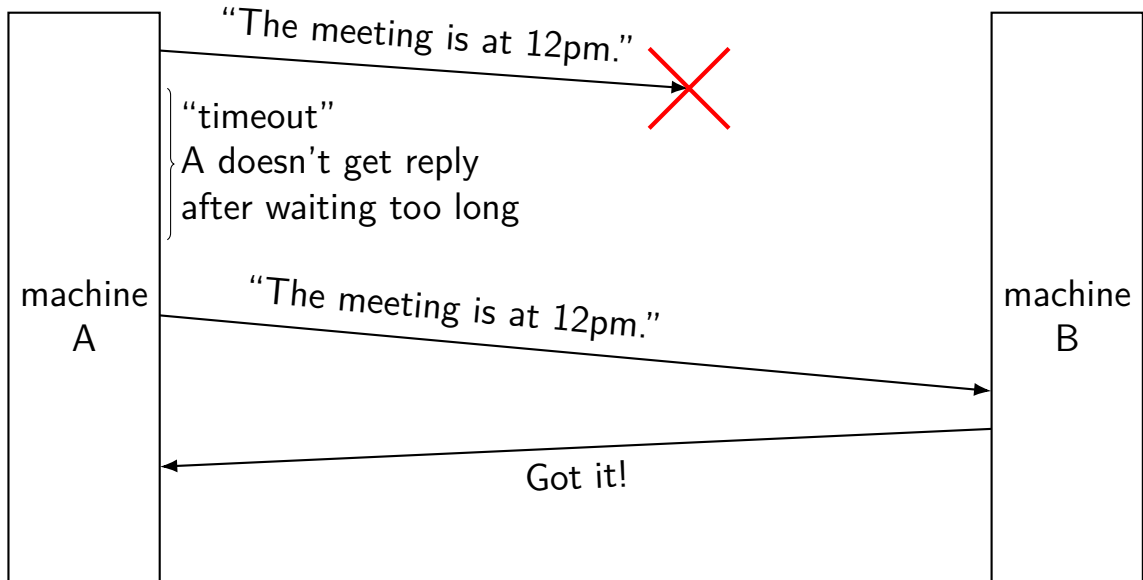
handling lost message



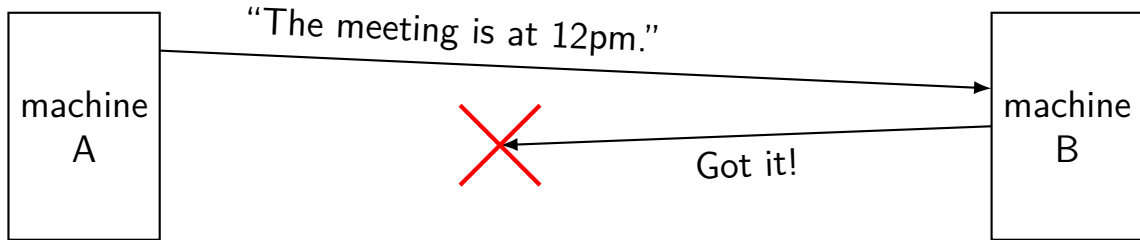
handling lost message



handling lost message



exercise: lost acknowledgement



exercise: how to fix this?

- A. machine A needs to send "Got 'got it!' "
- B. machine B should resend "Got it!" on its own
- C. machine A should resend the original message on its own
- D. none of these

answers

send “Got ‘got it!’ ”?

same problem: Now send ‘Got Got Got it’?

resend “Got it!” own its own?

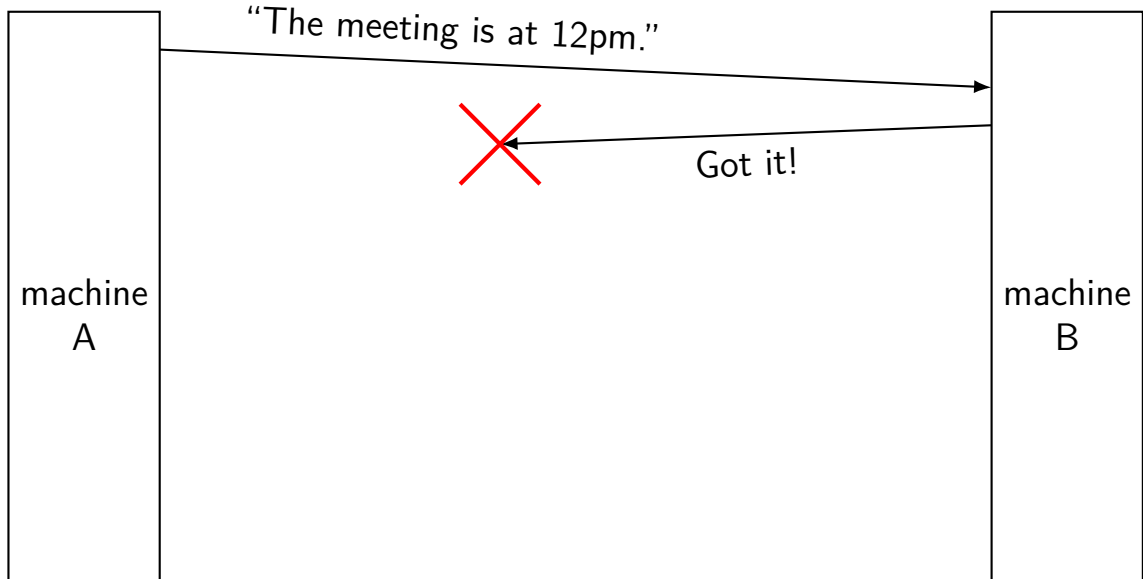
how many times? — B doesn't have that info

resend original message?

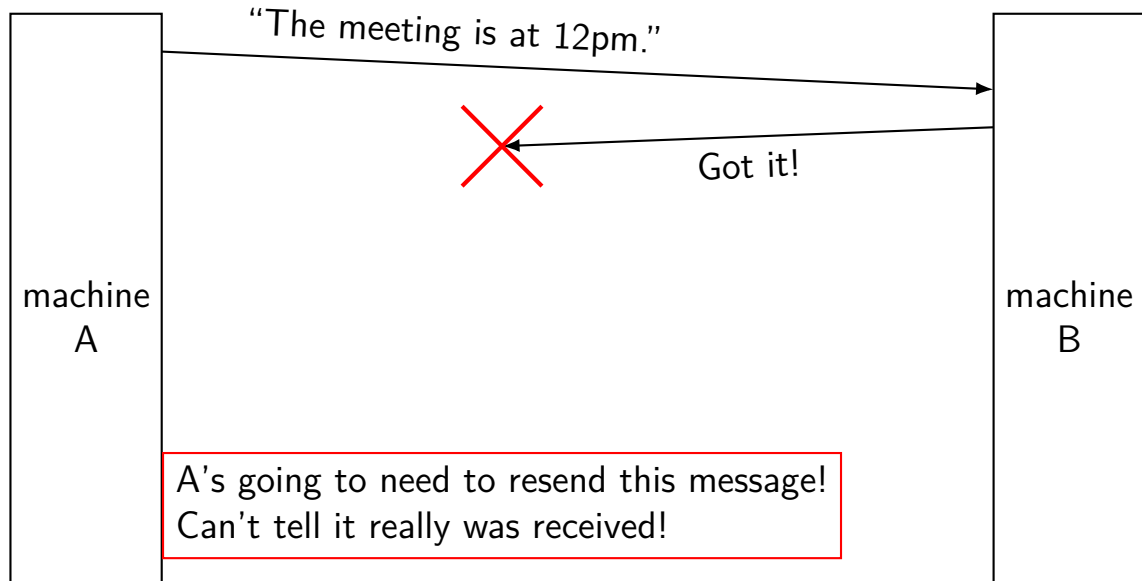
yes!

as far as machine A can see, *exact same situation* as losing original message

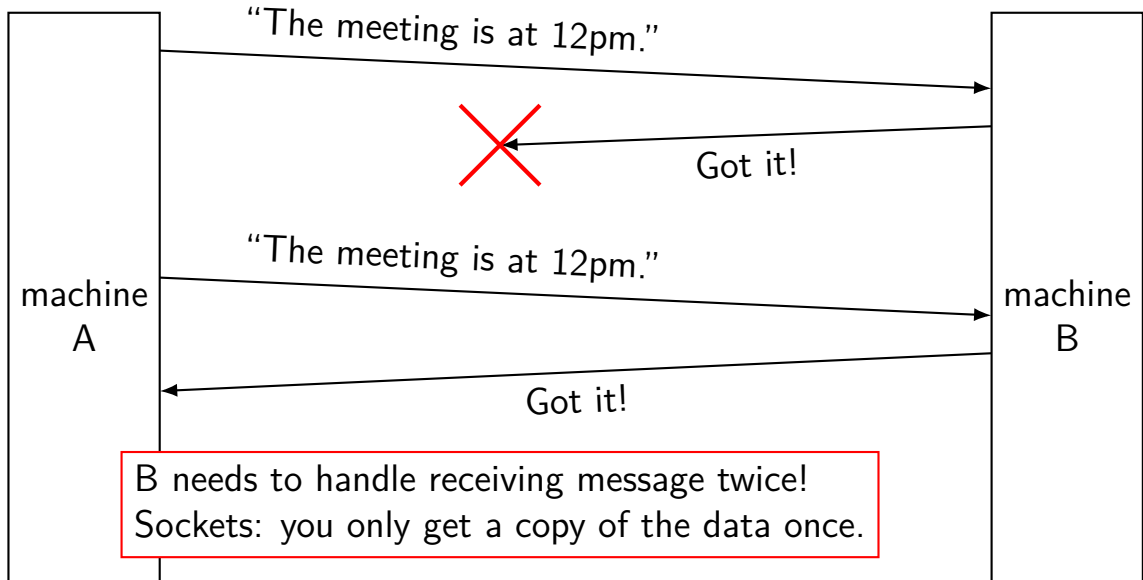
lost acknowledgements



lost acknowledgements



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network limitations/failures

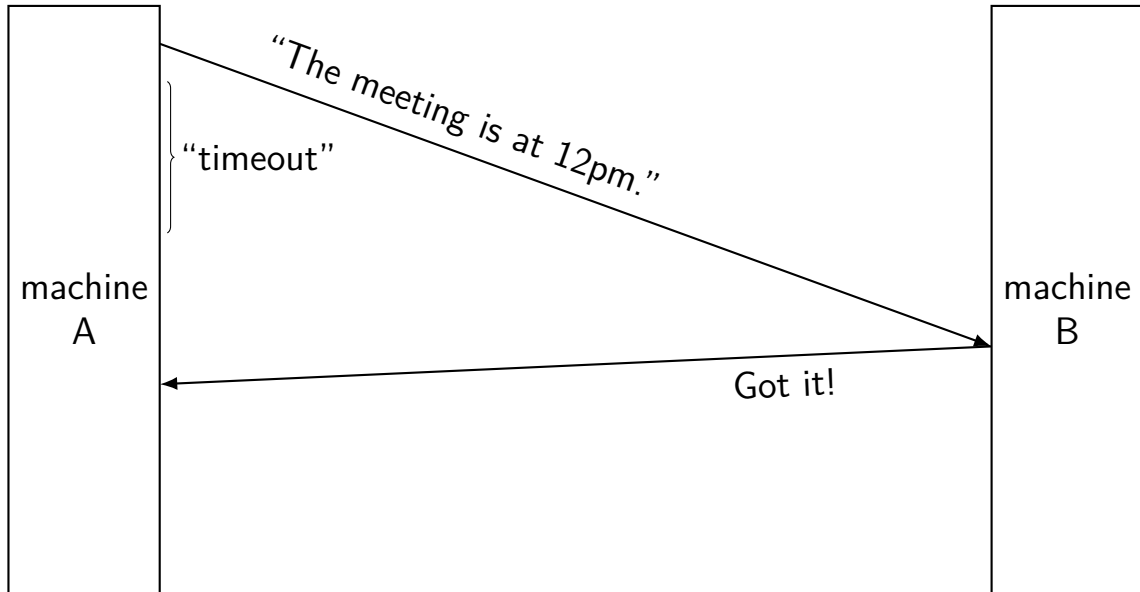
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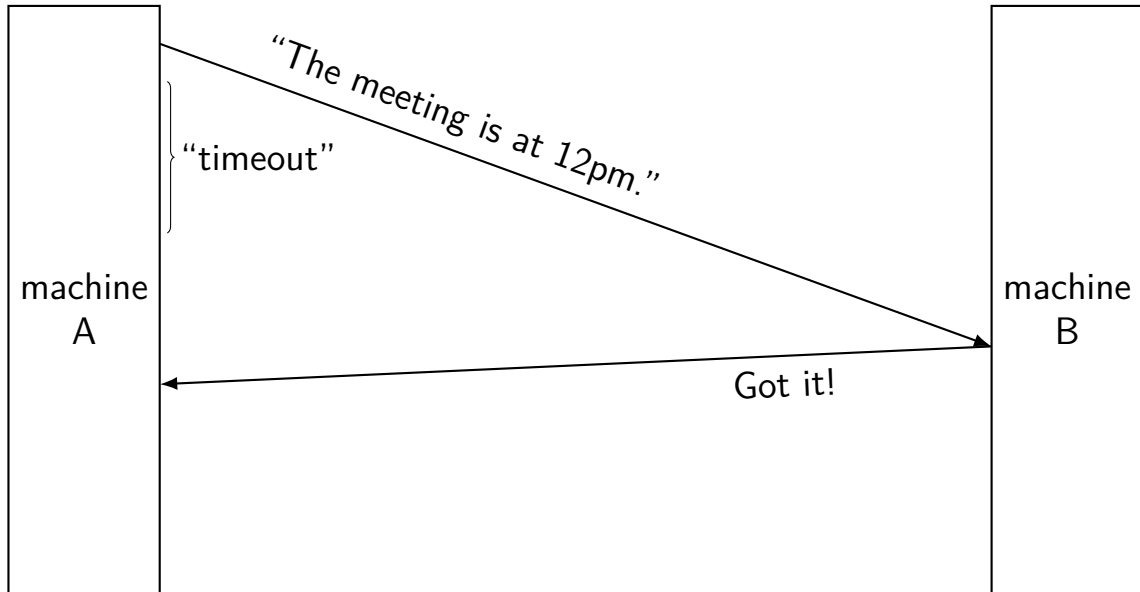
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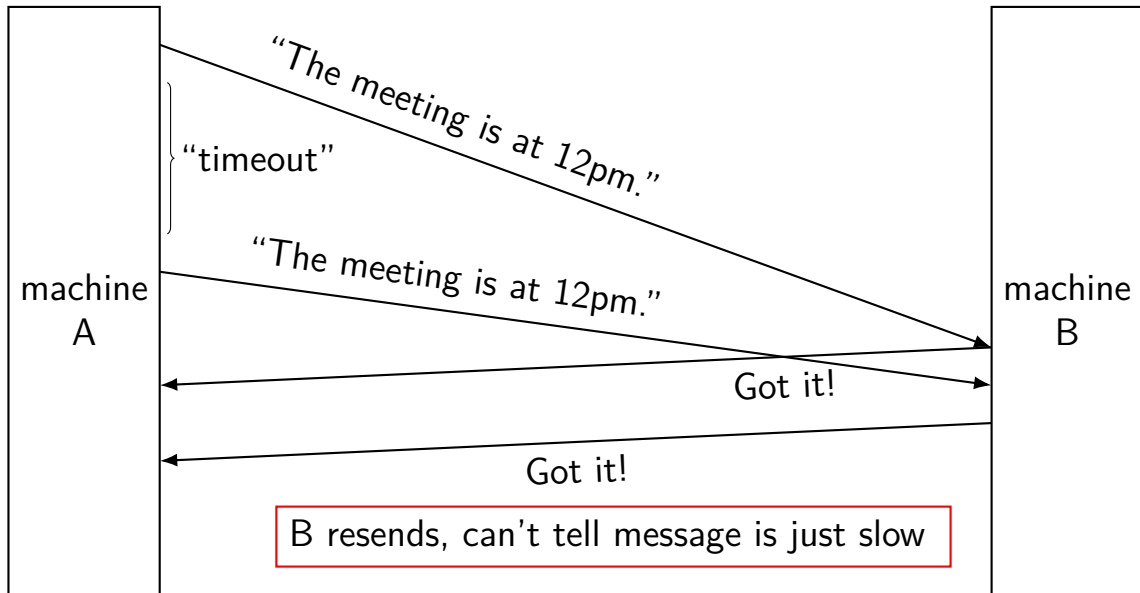
delayed message



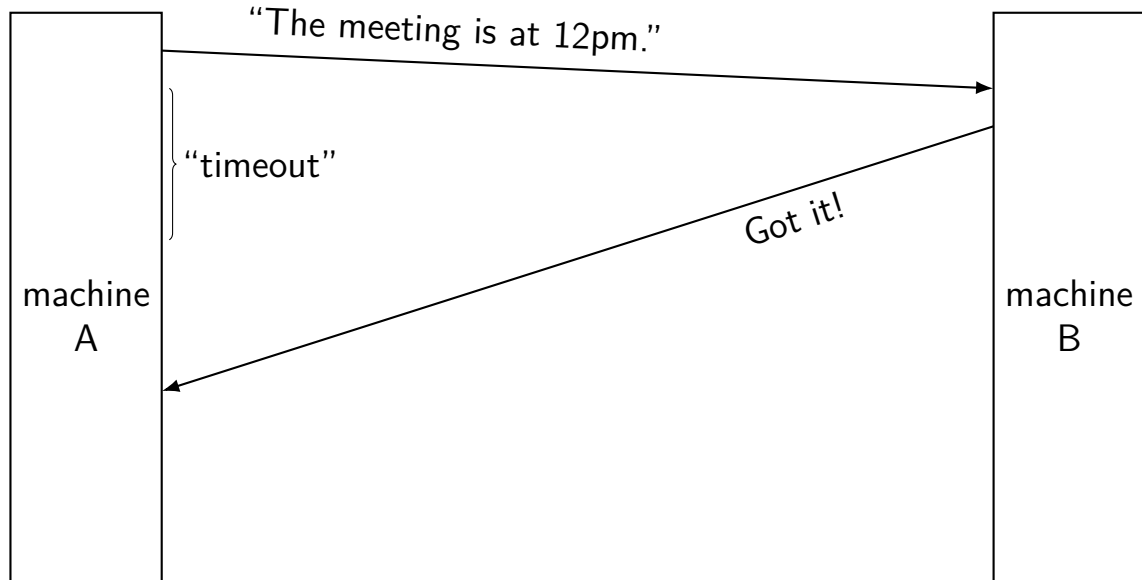
delayed message



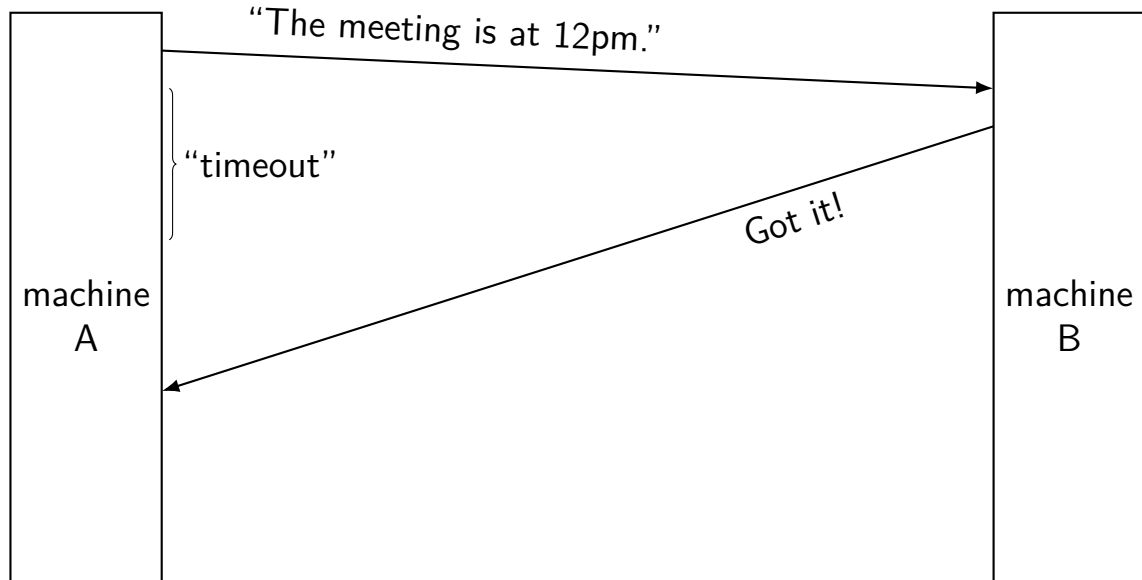
delayed message



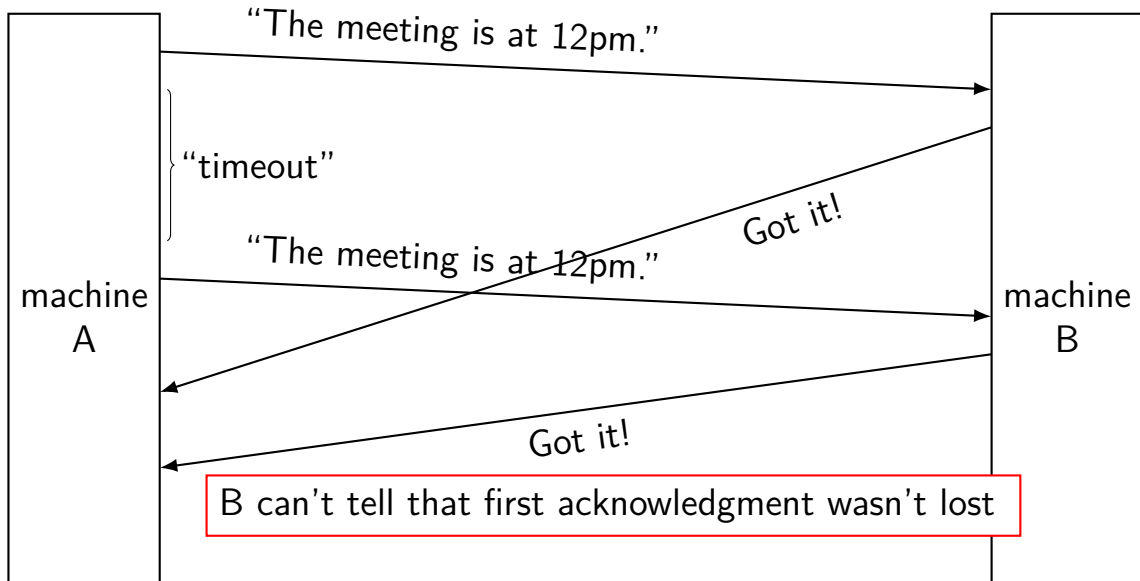
delayed acknowledgements



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network limitations/failures

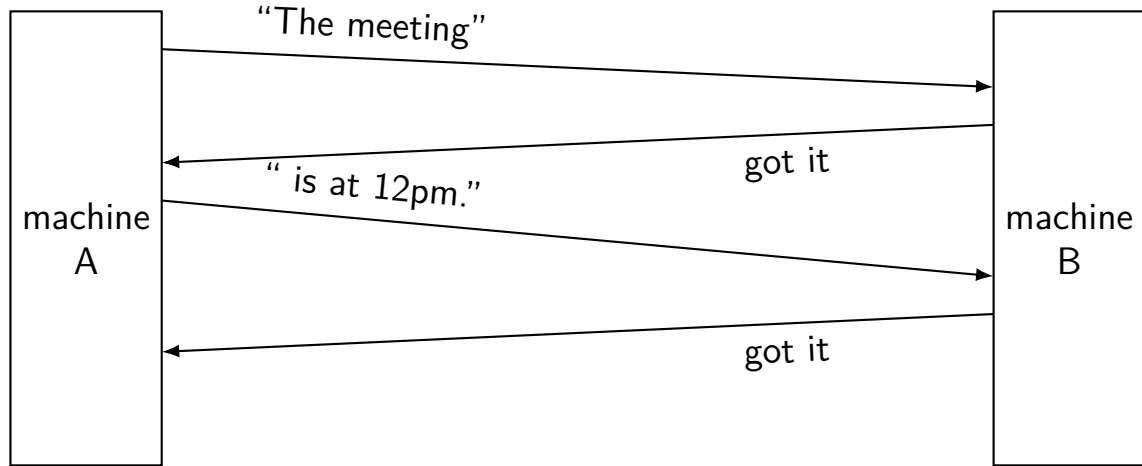
messages lost

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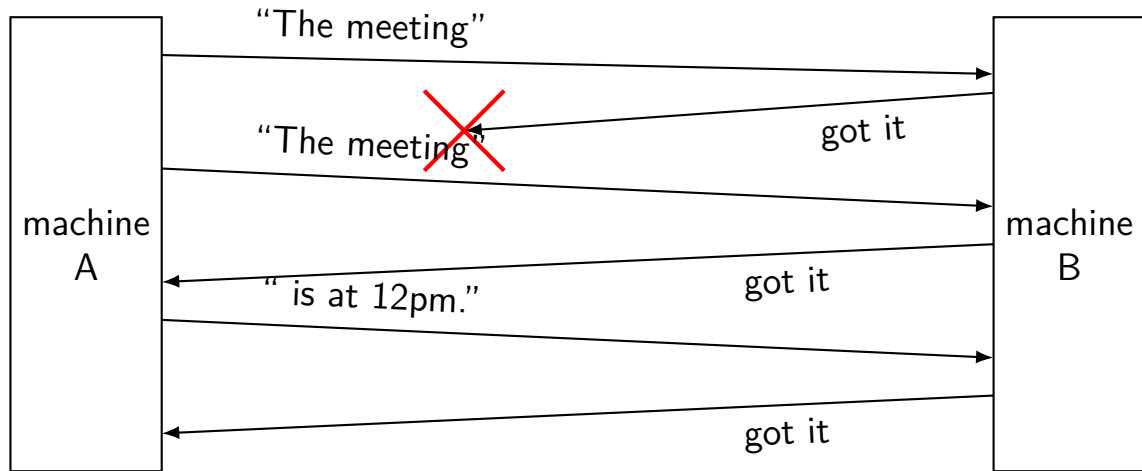
messages corrupted

splitting messages: try 1

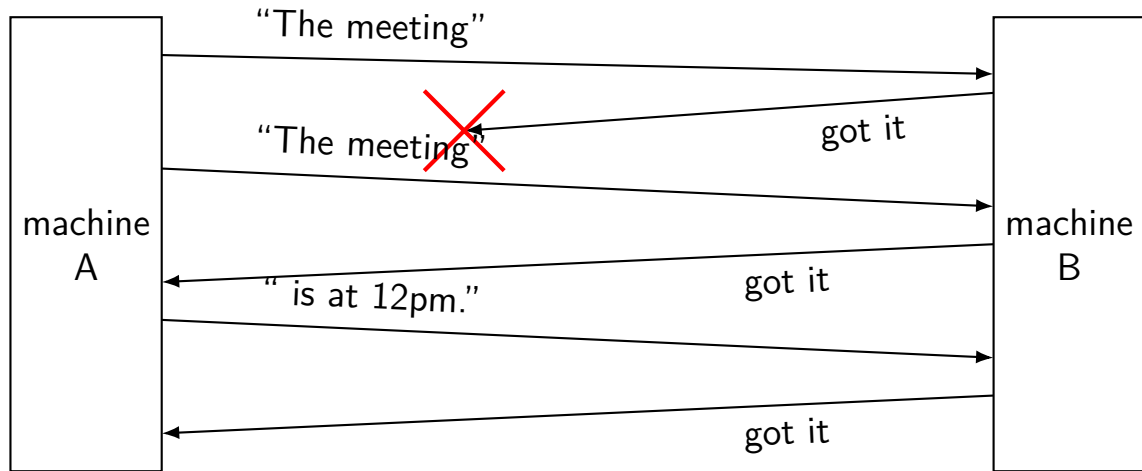


reconstructed message:
The meeting is at 12pm.

splitting messages: try 1 — problem 1



splitting messages: try 1 — problem 1



reconstructed message:

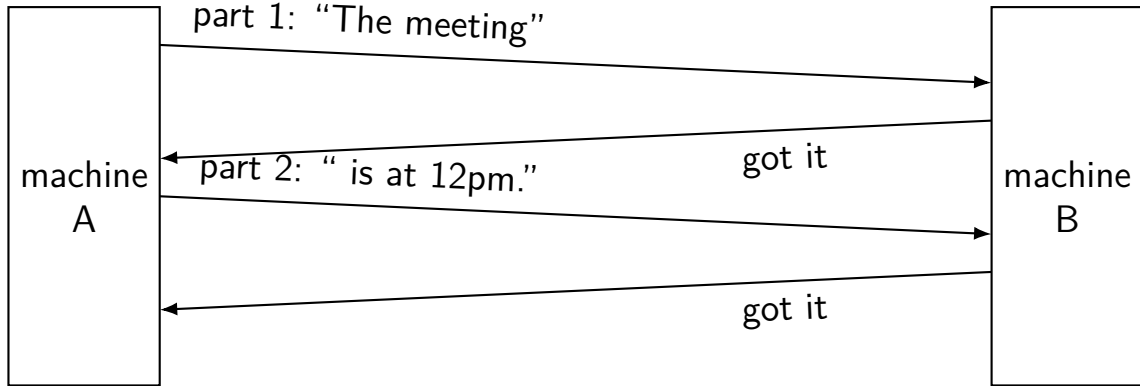
The meetingThe meeting is at 12pm.

exercise: other problems?

other scenarios where we'd also have problems?

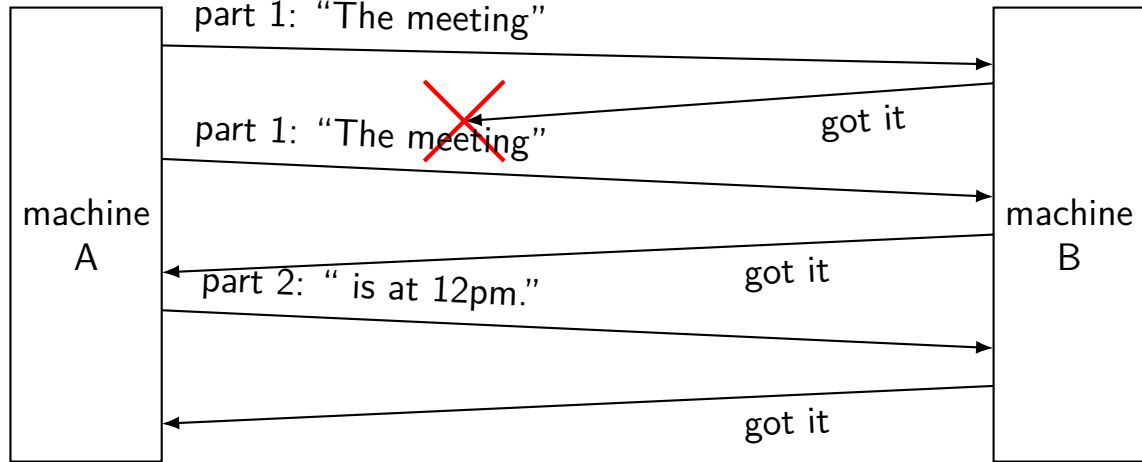
1. message (instead of acknowledgment) is lost
2. first message from machine A is delayed a long time by network
3. acknowledgment of second message lost instead of first

splitting messages: try 2



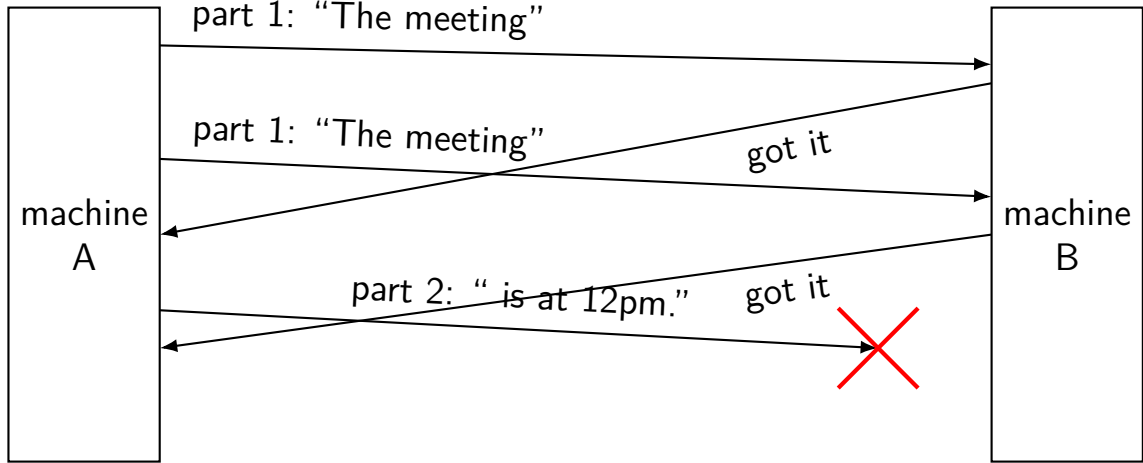
reconstructed message:
The meeting is at 12pm.

splitting messages: try 2 — missed ack



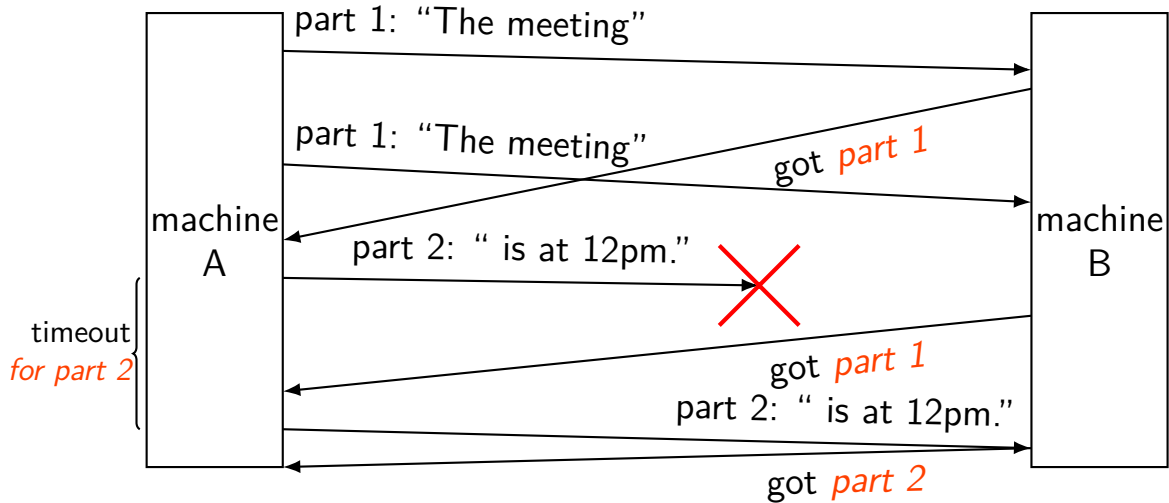
reconstructed message:
The meeting is at 12pm.

splitting messages: try 2 — problem



A thinks: part 1 + part 2 acknowledged!

splitting messages: version 3



network limitations/failures

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messages delayed/reordered

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messages corrupted

message corrupted

corruption: e.g., a bit flip

sent: I LIKE CATS (49204C494B452043415453)

recv: I LIKE BATS (49204C494B452042415453)

instead of sending “message”, send “message” + checksum

checksum is some calculation using the original message

ex: $\text{checksum}(\text{I LIKE CATS}) = 0xD9$

send 49204C494B452043415453D9

receiver then also computes the checksum with the same data

if matches: keep the message

if does not match: pretend like the message was lost (i.e., drop the message)

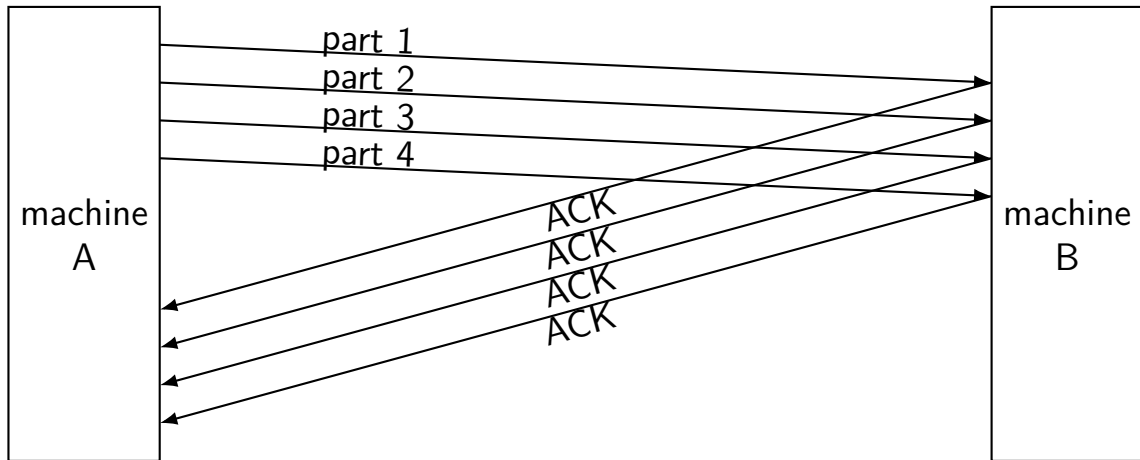
going faster

so far: send one message, wait for acknowledgment

very slow!

instead, can send a bunch of parts and get them acknowledged together

transmission window (ex: size 4)



Send a *window* of parts speculatively, then wait for ACKs.

transport layer protocols: UDP vs. TCP

TCP: stream to other program

- reliable transmission of as much data as you want

- “connecting” fails if server not responding

- `write(fd, "a", 1); write(fd, "b", 1) = write(fd, "ab", 2)`

- (at least) one socket per remote program being talked to

UDP: messages sent to program, but no reliability/streams

- unreliable transmission of short messages

- `write(fd, "a", 1); write(fd, "b", 1)  $\neq$  write(fd, "ab", 2)`

- “connecting” just sets default destination

- can `sendto()/recvfrom()` multiple other programs with one socket

- (but don't have to)

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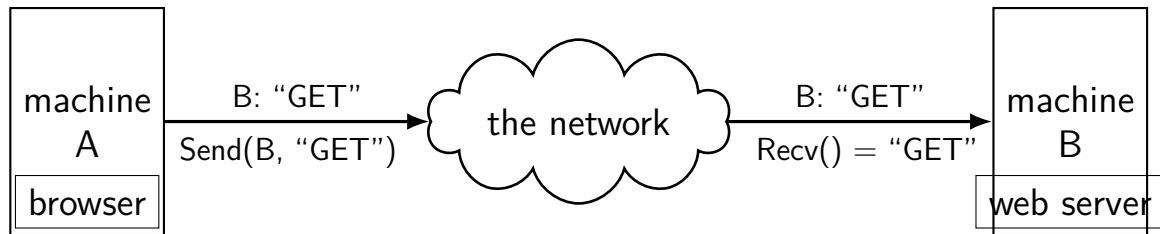
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networking layers

application	HTTP, SSH, SMTP, ...	application-defined meanings
transport	TCP, UDP, ...	reach correct program, reliability/streams
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link	Ethernet, Wi-Fi, ...	coordinate shared wire/radio
physical	Ethernet, Wi-Fi, ...	encode bits for wire/radio

mailbox model: importance of naming

mailbox abstraction: send/receive messages



how does the network know what "B" is?
how does B know the message is for the web server?
how does the response get to "A"?

names and addresses

name	address
<i>logical identifier</i>	<i>location/how to locate</i>
variable counter	memory address 0x7FFF9430
DNS name www.virginia.edu	IPv4 address 128.143.22.36
DNS name mail.google.com	IPv4 address 216.58.217.69
DNS name mail.google.com	IPv6 address 2607:f8b0:4004:80b::2005
DNS name reiss-t3620.cs.virginia.edu	IPv4 address 128.143.67.91
DNS name reiss-t3620.cs.virginia.edu	MAC address 18:66:da:2e:7f:da
service name https	port number 443
service name ssh	port number 22

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the network layer

the Internet Protocol (IP) version 4 or version 6

there are also others, but quite uncommon today

allows send messages to/recv messages from other networks

“internetwork”

messages usually called “packets”

IPv4 addresses

32-bit numbers

typically written like 128.143.67.11

four 8-bit decimal values separated by dots

first part is most significant

same as $128 \cdot 256^3 + 143 \cdot 256^2 + 67 \cdot 256 + 11 = 2\,156\,782\,459$

organizations get blocks of IPs

e.g. UVA has 128.143.0.0–128.143.255.255

e.g. Google has 216.58.192.0–216.58.223.255 and

74.125.0.0–74.125.255.255 and 35.192.0.0–35.207.255.255

some IPs reserved for non-Internet use (127.*, 10.*, 192.168.*)

IPv6 addresses

IPv6 like IPv4, but with 128-bit numbers

written in hex, 16-bit parts, separated by colons (:)

strings of 0s represented by double-colons (::)

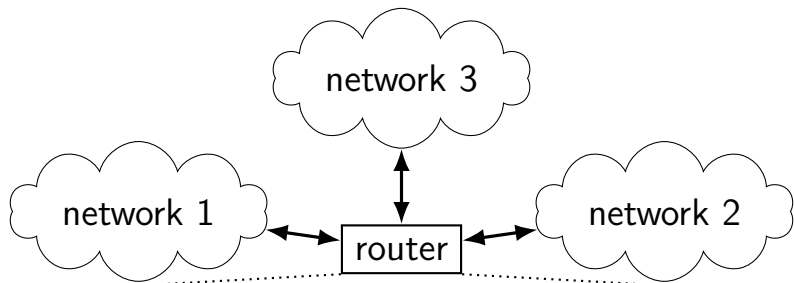
typically given to users in blocks of 2^{80} or 2^{64} addresses
no need for address translation?

`2607:f8b0:400d:c00::6a =`

`2607:f8b0:400d:0c00:0000:0000:0000:006a`

`2607f8b0400d0c000000000000000000006a`_{SIXTEEN}

IPv4 addresses and routing tables



if I receive data for...	send it to...
128.143.0.0—128.143.255.255	network 1, 11.4.3.2
192.107.102.0—192.107.102.255	network 1, 11.4.3.2
...	...
4.0.0.0—7.255.255.255	network 2, 12.4.6.4
64.8.0.0—64.15.255.255	network 2, 45.4.0.1
...	...
anything else	network 3, 199.44.33.1

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port numbers

we run multiple programs on a machine

IP addresses identifying machine — not enough

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so, add 16-bit *port numbers*

think: multiple PO boxes at address

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0–49151: typically assigned for particular services

80 = http, 443 = https, 22 = ssh, ...

49152–65535: allocated on demand

default “return address” for client connecting to server

connections in TCP/IP

connection identified by *4-tuple*

used by OS to lookup “where is the socket?”

(local IP address, local port, remote IP address, remote port)

local IP address, port number can be set with `bind()` function

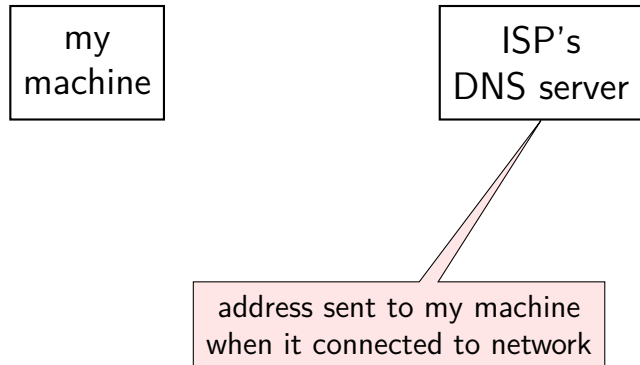
typically always done for servers, not done for clients

system will choose default if you don't

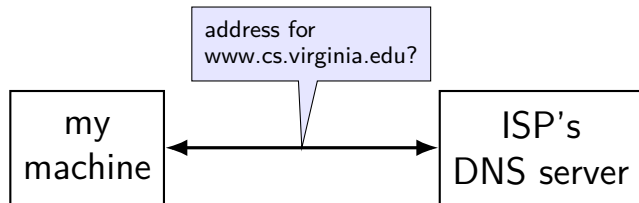
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DNS name reiss-t3620.cs.virginia.edu	IPv4 address 128.143.67.91
DNS name reiss-t3620.cs.virginia.edu	MAC address 18:66:da:2e:7f:da
service name https	port number 443
service name ssh	port number 22

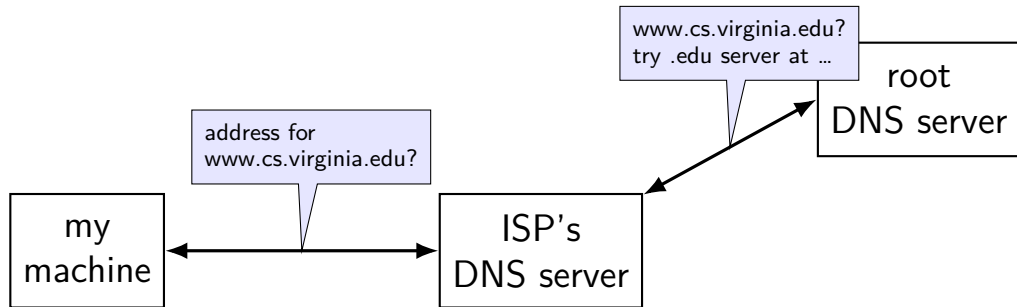
DNS: Domain Name System



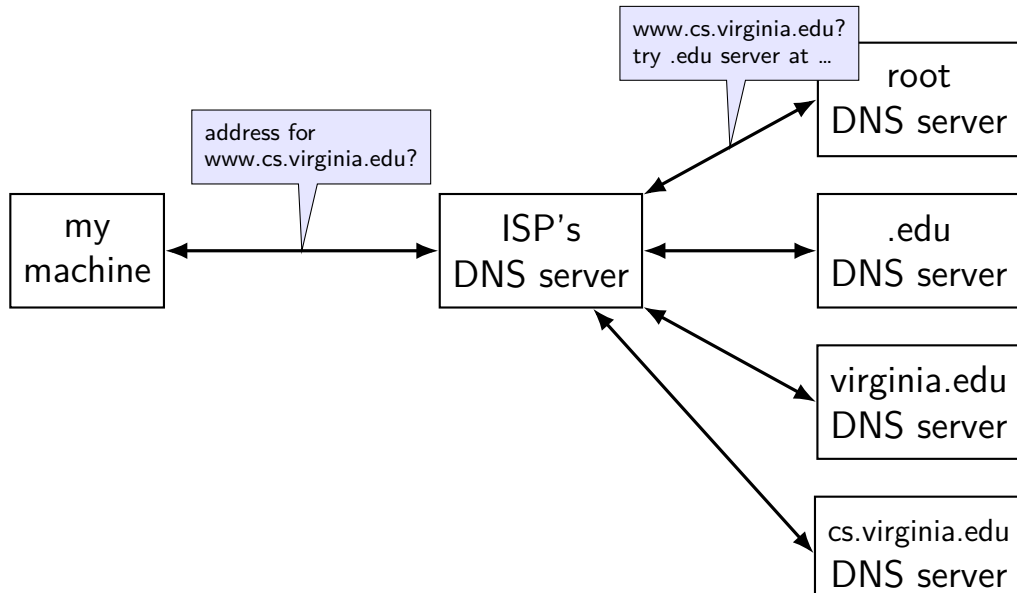
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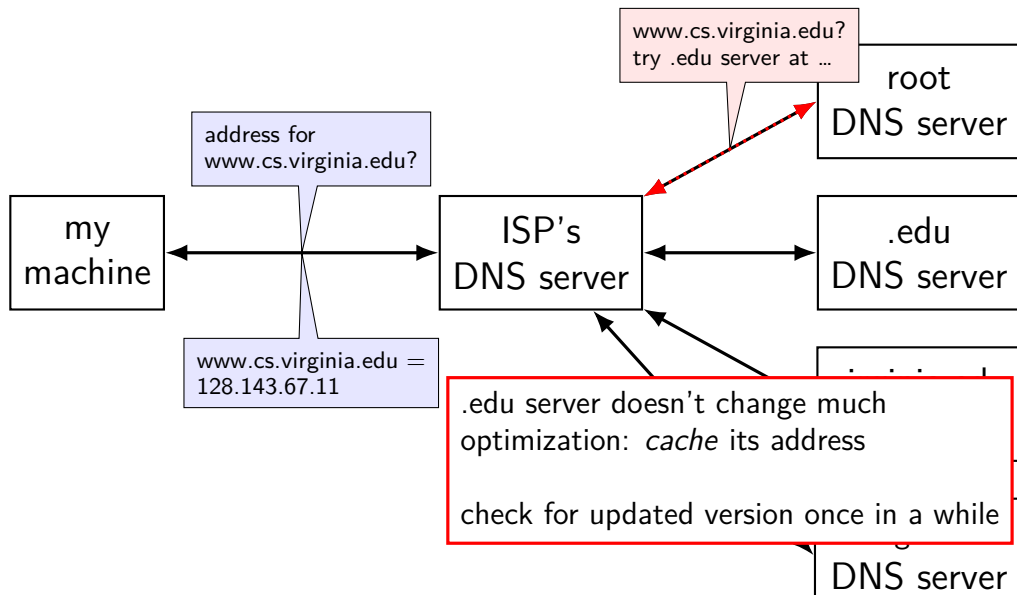
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URL / URIs

Uniform Resource Locators (URL)

tells how to find “resource” on network

uniform — one syntax for multiple protocols (types of servers, etc.)

Uniform Resource Identifiers

superset of URLs

URI examples

`https://kytos02.cs.virginia.edu:443/cs3130-spring2023/
quizzes/quiz.php?qid=02#q2`

`https://kytos02.cs.virginia.edu/cs3130-spring2023/
quizzes/quiz.php?qid=02`

`https://www.cs.virginia.edu/`

`sftp://cr4bd@portal.cs.virginia.edu/u/cr4bd/file.txt`

`tel:+1-434-982-2200`

`//www.cs.virginia.edu/~cr4bd/3130/S2023/
/~cr4bd/3130/S2023`

scheme and/or host implied from context

URI generally

scheme://authority/path?query#fragment

scheme: — what protocol

//authority/

authorirty = user@host:port OR host:port OR user@host OR host

path

which resource

?query — usually key/value pairs

#fragment — place in resource

most components (sometimes) optional

HTTP: HyperText Transfer Protocol

primary application-layer protocol for the Web

standard port: 80

uses TCP

client-server protocol

ex. URL: `http://www.foo.com/bar`

clients make requests to URLs

common HTTP commands: GET, HEAD, POST

servers respond with status and data

common status codes: 200 OK, 404 NOT FOUND, 500 ERR

why IPv4 *and* IPv6?

ran out of IPv4 addresses

2^{32} *seemed* like a lot of addresses

now there are more internet *users* than addresses

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possible solutions:

1. IPv6: use more bits for addresses, enough to never run out

why IPv4 *and* IPv6?

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2^{32} *seemed* like a lot of addresses

now there are more internet *users* than addresses

possible solutions:

1. IPv6: use more bits for addresses, enough to never run out
2. NAT: use one IPv4 address for an entire network of devices

NAT: network address translation

convert many private addrs. to one public addr.

locally: use private IP addresses for local addresses

outside: private IP addresses become a single public one

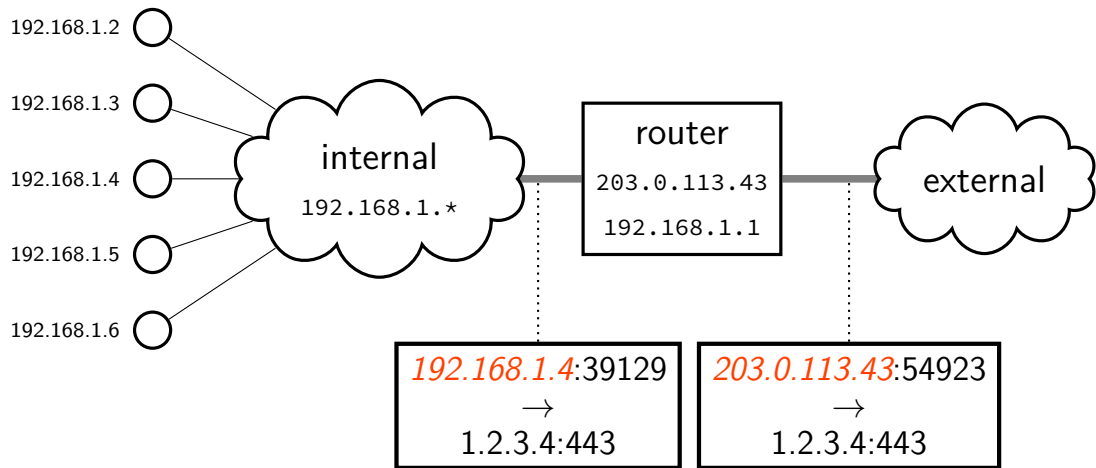
commonly how home networks work (and some ISPs)

certain IPv4 address blocks reserved for private/internal use

192.168.X.X

172.16.X.X

NAT idea



NAT illusion

NAT illusion:

private IP address communicating directly with public IP

inside network, talking to outside:

- use private local address
- use public remote address
- never see router's address

outside network, talking to inside

- use public local address
- use router's public address

implementing NAT

remote host + port	outside local port number	inside IP	inside port number
128.148.17.3:443	54033	192.168.1.5	43222
11.7.17.3:443	53037	192.168.1.5	33212
128.148.31.2:22	54032	192.168.1.37	43010
128.148.17.3:443	63039	192.168.1.37	32132

table of the translations

need to update as new connections made

upcoming lab

request + receive message split into pieces

you are responsible for:

- requesting parts in order

- resending requests if messages lost/corrupted

“acknowledge” receiving part X to request part $X+1$

upcoming lab

request + receive message split into pieces

you are responsible for:

- requesting parts in order

- resending requests if messages lost/corrupted*

- “acknowledge” receiving part X to request part $X+1$*

protocol

GET x — retrieve message x ($x = 0, 1, 2$, or 3)

- other end acknowledges by giving data

- if they don't reply, you need to send again

- higher numbered messages have errors/etc. that are harder to handle

ACK n

- request message $n + 1$ by acknowledging message n

- not quite same purpose as acknowledgments in prior examples

- (in lab, the response is your 'acknowledgment' of your request;

- you retry if you don't get it)

protocol

GET x — retrieve message x ($x = 0, 1, 2$, or 3)

other end acknowledges by giving data

if they don't reply, you need to send again

higher numbered messages have errors/etc. that are harder to handle

ACK n

request message $n + 1$ by acknowledging message n

not quite same purpose as acknowledgments in prior examples

*(in lab, the response is your 'acknowledgment' of your request;
you retry if you don't get it)*

callback-based programming (1)

```
/* library code you don't write */
/* lab: part of waitForAllTimeoutsAndMessagesThenExit() */
void mainLoop() {
    while (notExiting) {
        Event event = waitForAndGetNextEvent();
        if (event.type == RECIEVED) {
            recvd(...);
        } else if (event.type == TIMEOUT) {
            (event.timeout_function)(...);
        }
        ...
    }
}
```

callback-based programming (2)

```
/* your code, called by library */
void recvd(...) {
    ...
    setTimeout(..., timerCallback, ...);
}

void timerCallback(...) {
    ...
}

int main() {
    send(.../* first message */);
    ... /* other initial setup */
    waitForAllTimeoutsAndMessagesThenExit(); // runs mainLoop
}
```


callback-based programming

writing scripts in a webpage

many graphical user interface libraries

sometimes servers that handle lots of connections

backup slides

the link layer

Ethernet, Wi-Fi, Bluetooth, DOCSIS (cable modems), ...

allows send/recv messages to machines on “same” network segment

- typically: wireless range+channel or connected to a single switch/router
- could be larger (if *bridging* multiple network segments)
- could be smaller (switch/router uses “virtual LANs”)

typically: source+destination specified with MAC addresses

- MAC = media access control

- usually manufacturer assigned / hard-coded into device
- unique address per port/wifi transmitter/etc.

can specify destination of “anyone” (called *broadcast*)

the link layer

Ethernet, Wi-Fi, Bluetooth, DOCSIS (cable modems), ...

allows send/recv messages to machines on “*same*” *network segment*

- typically: wireless range+channel or connected to a single switch/router
- could be larger (if *bridging* multiple network segments)
- could be smaller (switch/router uses “virtual LANs”)

typically: source+destination specified with MAC addresses

- MAC = media access control

- usually manufacturer assigned / hard-coded into device

- unique address per port/wifi transmitter/etc.

can specify destination of “anyone” (called *broadcast*)

link layer jobs

divide raw bits into messages

identify who message is for on shared radio/wire

handle if two+ machines use radio/wire at same time

drop/resend messages if corruption detected

resending more common in radio schemes (wifi, etc.)

link layer reliability?

Ethernet + Wifi have checksums

Q1: Why doesn't this give us uncorrupted messages?

Why do we still have checksums at the higher layers?

Q2: What's a benefit of doing this if we're also doing it in the higher layer?

link layer quality of service

if frame gets...

event	on Ethernet	on WiFi
collides with another	detected + may resend	resend
not received	lose silently	resent
header corrupted	usually discard silently	usually resend
data corrupted	usually discard silently	usually resend
too long	not allowed to send	not allowed to send
reordered (v. other messages)	received out of order	received out of order
destination unknown	lose silently	usually resend??
too much being sent	discard excess?	discard excess?

network layer quality of service

if packet ...

event	on IPv4/v6
collides with another	out of scope — handled by link layer
not received	lost silently
header corrupted	usually discarded silently
data corrupted	received corrupted
too long	dropped with notice or “fragmented” + recombined
reordered (v. other messages)	received out of order
destination unknown	usually dropped with notice
too much being sent	discard excess

network layer quality of service

if packet ...

event	on IPv4/v6
collides with another	out of scope — handled by link layer
<i>not received</i>	lost silently
header corrupted	usually discarded silently
data corrupted	received corrupted
too long	dropped with notice or “fragmented” + recombined
reordered (v. other messages)	received out of order
destination unknown	usually dropped with notice
too much being sent	discard excess

includes dropped by link layer
(e.g. if detected corrupted there)

firewalls

don't want to expose network service to everyone?

solutions:

- service picky about who it accepts connections from
- filters in OS on machine with services
- filters on router

later two called “firewalls”

firewall rules examples?

ALLOW tcp port 443 (https) FROM everyone

ALLOW tcp port 22 (ssh) FROM *my desktop's IP address*

BLOCK tcp port 22 (ssh) FROM everyone else

ALLOW from address X to address Y

...

TCP state machine

TIME_WAIT, ESTABLISHED, ...?

OS tracks “state” of TCP connection

- am I just starting the connection?

- is other end ready to get data?

- am I trying to close the connection?

- do I need to resend something?

standardized set of state names

TIME_WAIT

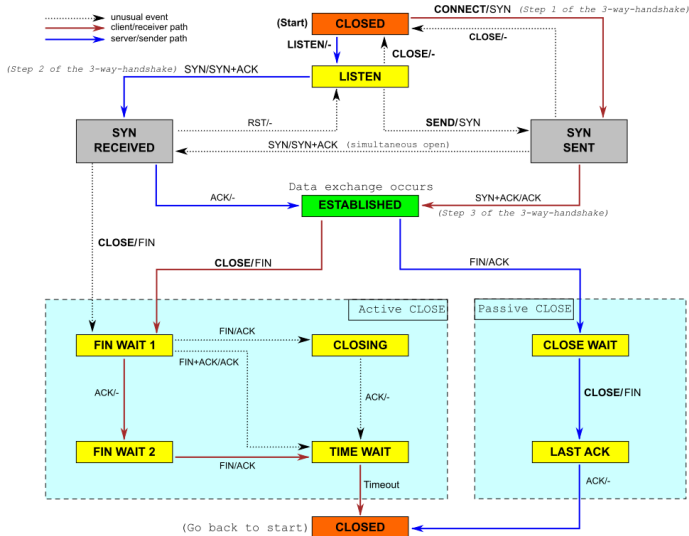
remember delayed messages?

problem for TCP ports

if I reuse port number, I can get message from old connection

solution: TIME_WAIT to make sure connection really done
done after sending last message in connection

TCP state machine picture



querying the root

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
edu.                172800      IN          NS          b.edu-servers.net.
edu.                172800      IN          NS          f.edu-servers.net.
edu.                172800      IN          NS          i.edu-servers.net.
edu.                172800      IN          NS          a.edu-servers.net.
...
b.edu-servers.net.  172800      IN          A           191.33.14.30
b.edu-servers.net.  172800      IN          AAAA        2001:503:231d::2:30
f.edu-servers.net.  172800      IN          A           192.35.51.30
f.edu-servers.net.  172800      IN          AAAA        2001:503:d414::30
...
;; Received 843 bytes from 198.97.190.53#53(h.root-servers.net) in 8 ms
...
```

querying the edu

```
$ dig +trace +all www.cs.virginia.edu
```

```
...  
virginia.edu.          172800      IN          NS          nom.virginia.edu.  
virginia.edu.          172800      IN          NS          uvaarpa.virginia.edu.  
virginia.edu.          172800      IN          NS          eip-01-aws.net.virginia.edu.  
nom.virginia.edu.      172800      IN          A           128.143.107.101  
uvaarpa.virginia.edu.  172800      IN          A           128.143.107.117  
eip-01-aws.net.virginia.edu. 172800 IN          A           44.234.207.10  
;; Received 165 bytes from 192.26.92.30#53(c.edu-servers.net) in 40 ms  
...
```


querying virginia.edu+cs.virginia.edu

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
```

```
cs.virginia.edu.          3600      IN      NS      coresrv01.cs.virginia.edu.
```

```
coresrv01.cs.virginia.edu. 3600      IN      A      128.143.67.11
```

```
;; Received 116 bytes from 44.234.207.10#53(eip-01-aws.net.virginia.edu) in 72 ms
```

```
www.cs.Virginia.EDU.      172800   IN      A      128.143.67.11
```

```
cs.Virginia.EDU.          172800   IN      NS      coresrv01.cs.Virginia.EDU.
```

```
coresrv01.cs.Virginia.EDU. 172800   IN      A      128.143.67.11
```

```
;; Received 151 bytes from 128.143.67.11#53(coresrv01.cs.virginia.edu) in 4 ms
```

querying typical ISP's resolver

```
$ dig www.cs.virginia.edu
...
;; ANSWER SECTION:
www.cs.Virginia.EDU.      7183      IN      A      128.143.67.11
..
```

cached response

valid for 7183 more seconds

after that everyone needs to check again

'connected' UDP sockets

```
int fd = socket(AF_INET, SOCK_DGRAM, 0);
struct sockaddr_in my_addr= ...;
/* set local IP address + port */
bind(fd, &my_addr, sizeof(my_addr))
struct sockaddr_in to_addr = ...;
connect(fd, &to_addr); /* set remote IP address + port */
/* doesn't actually communicate with remote address yet */
...
int count = write(fd, data, data_size);
// OR
int count = send(fd, data, data_size, 0 /* flags */);
/* single message -- sent ALL AT ONCE */

int count = read(fd, buffer, buffer_size);
// OR
int count = recv(fd, buffer, buffer_size, 0 /* flags */);
/* receives whole single message ALL AT ONCE */
```

UDP sockets on IPv4

```
int fd = socket(AF_INET, SOCK_DGRAM, 0);
struct sockaddr_in my_addr= ...;
/* set local IP address + port */
if (0 != bind(fd, &my_addr, sizeof(my_addr)))
    handle_error();

...
struct sockaddr_in to_addr = ...;
/* send a message to specific address */
int bytes_sent = sendto(fd, data, data_size, 0 /* flags */,
    &to_addr, sizeof(to_addr));

struct sockaddr_in from_addr = ...;
/* receive a message + learn where it came from */
int bytes_rcvd = recvfrom(fd, &buffer[0], buffer_size, 0,
    &from_addr, sizeof(from_addr));

...
```

what about non-local machines?

when configuring network specify:

range of addresses to expect on local network

128.148.67.0-128.148.67.255 on my desktop

“netmask”

gateway machine to send to for things outside my local network

128.143.67.1 on my desktop

my desktop looks up the corresponding MAC address

routes on my desktop

```
$ /sbin/route -n
```

```
Kernel IP routing table
```

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
0.0.0.0	128.143.67.1	0.0.0.0	UG	100	0	0	enp0s31f6
128.143.67.0	0.0.0.0	255.255.255.0	U	100	0	0	enp0s31f6
169.254.0.0	0.0.0.0	255.255.0.0	U	1000	0	0	enp0s31f6

network configuration says:

(line 2) to get to 128.143.67.0–128.143.67.255, send directly on local network

“genmask” is mask (for bitwise operations) to specify how big range is

(line 3) to get to 169.254.0.0–169.254.255.255, send directly on local network

(line 1) to get anywhere else, use “gateway” 128.143.67.1

names and addresses

name	address
<i>logical identifier</i>	<i>location/how to locate</i>
variable counter	memory address 0x7FFF9430
DNS name www.virginia.edu	IPv4 address 128.143.22.36
DNS name mail.google.com	IPv4 address 216.58.217.69
DNS name mail.google.com	IPv6 address 2607:f8b0:4004:80b::2005
DNS name reiss-t3620.cs.virginia.edu	IPv4 address 128.143.67.91
DNS name reiss-t3620.cs.virginia.edu	MAC address 18:66:da:2e:7f:da
service name https	port number 443
service name ssh	port number 22

two types of addresses?

MAC addresses: on link layer

IP addresses: on network layer

how do we know which MAC address to use?

a table on my desktop

my desktop:

```
$ arp -an
? (128.143.67.140) at 3c:e1:a1:18:bd:5f [ether] on enp0s31f6
? (128.143.67.236) at <incomplete> on enp0s31f6
? (128.143.67.11) at 30:e1:71:5f:39:10 [ether] on enp0s31f6
? (128.143.67.92) at <incomplete> on enp0s31f6
? (128.143.67.5) at d4:be:d9:b0:99:d1 [ether] on enp0s31f6
```

...

network address to link-layer address + interface

only tracks things directly connected to my local network

non-local traffic sent to local router

how is that table made?

ask all machines on local network (same switch)

“Who has 128.148.67.140”

the correct one replies

URLs and HTTP (1)

`http://www.foo.com:80/foo/bar?quux#q1`

lookup IP address of `www.foo.com`

connect via TCP to port 80:

`GET /foo/bar?quux HTTP/1.1`

`Host: www.foo.com:80`

URLs and HTTP (1)

`http://www.foo.com:80/foo/bar?quux#q1`

lookup IP address of `www.foo.com`

connect via TCP to port 80:

`GET /foo/bar?quux HTTP/1.1`

`Host: www.foo.com:80`

URLs and HTTP (1)

`http://www.foo.com:80/foo/bar?quux#q1`

lookup IP address of `www.foo.com`

connect via TCP to port 80:

`GET /foo/bar?quux HTTP/1.1`

`Host: www.foo.com:80`

exercise: why include the Host there?

querying the root

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
edu.                172800      IN          NS          b.edu-servers.net.
edu.                172800      IN          NS          f.edu-servers.net.
edu.                172800      IN          NS          i.edu-servers.net.
edu.                172800      IN          NS          a.edu-servers.net.
...
b.edu-servers.net.  172800      IN          A           191.33.14.30
b.edu-servers.net.  172800      IN          AAAA        2001:503:231d::2:30
f.edu-servers.net.  172800      IN          A           192.35.51.30
f.edu-servers.net.  172800      IN          AAAA        2001:503:d414::30
...
;; Received 843 bytes from 198.97.190.53#53(h.root-servers.net) in 8 ms
...
```

querying the edu

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
```

```
virginia.edu.          172800      IN          NS          nom.virginia.edu.  
virginia.edu.          172800      IN          NS          uvaarpa.virginia.edu.  
virginia.edu.          172800      IN          NS          eip-01-aws.net.virginia.edu.  
nom.virginia.edu.      172800      IN          A           128.143.107.101  
uvaarpa.virginia.edu.  172800      IN          A           128.143.107.117  
eip-01-aws.net.virginia.edu. 172800 IN          A           44.234.207.10  
;; Received 165 bytes from 192.26.92.30#53(c.edu-servers.net) in 40 ms  
...
```

querying virginia.edu+cs.virginia.edu

```
$ dig +trace +all www.cs.virginia.edu
```

```
...
```

```
cs.virginia.edu.          3600      IN      NS      coresrv01.cs.virginia.edu.
```

```
coresrv01.cs.virginia.edu. 3600      IN      A      128.143.67.11
```

```
;; Received 116 bytes from 44.234.207.10#53(eip-01-aws.net.virginia.edu) in 72 ms
```

```
www.cs.Virginia.EDU.      172800   IN      A      128.143.67.11
```

```
cs.Virginia.EDU.          172800   IN      NS      coresrv01.cs.Virginia.EDU.
```

```
coresrv01.cs.Virginia.EDU. 172800   IN      A      128.143.67.11
```

```
;; Received 151 bytes from 128.143.67.11#53(coresrv01.cs.virginia.edu) in 4 ms
```


querying typical ISP's resolver

```
$ dig www.cs.virginia.edu
```

```
...
```

```
;; ANSWER SECTION:
```

```
www.cs.Virginia.EDU.          7183          IN           A           128.143.67.11
```

```
..
```

cached response

valid for 7183 more seconds

after that everyone needs to check again

spoofing

if I only allow connections from my desktop's IP addresses,
how would you attack this?

hint: how do we know what address messages come from?

connection setup: server, manual

```
int server_socket_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY; /* "any address I can use" */
    /* or: addr.s_addr.in_addr = INADDR_LOOPBACK (127.0.0.1) */
    /* or: addr.s_addr.in_addr = htonl(...); */
addr.sin_port = htons(9999); /* port number 9999 */

if (bind(server_socket_fd, &addr, sizeof(addr)) < 0) {
    /* handle error */
}
listen(server_socket_fd, MAX_NUM_WAITING);
...
int socket_fd = accept(server_socket_fd, NULL);
```

connection setup: server, manual

```
int server_socket_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY; /* "any address I can use" */
    /* or: addr.sin_addr.in_addr = INADDR_LOOPBACK (127.0.0.1) */
    /* or: addr.sin_addr.in_addr = htonl(...); */
addr.sin_port = htons(9999); /* port number 9999 */

if (bind(server_socket_fd, &addr, sizeof(addr)) < 0) {
    /* handle error */
}

listen(server_socket_fd, 10);

...
int sock = accept(server_socket_fd, &addr, &addrlen);
```

INADDR_ANY: accept connections for any address I can!

alternative: specify specific address

connection setup: server, manual

```
int server_socket_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY; /* "any address I can use" */
/* or: addr.s_addr.in_addr = INADDR_LOOPBACK (127.0.0.1) */
/* or: addr.s_addr.in_addr = htonl(...); */
addr.sin_port = htons(9999); /* port number 9999 */

if (bind(server_socket_fd, &addr, sizeof(addr)) < 0) {
    /* handle error */
}
listen(server_socket_fd, 10);
int
```

bind to 127.0.0.1? only accept connections *from same machine*
what we recommend for FTP server assignment

connection setup: server, manual

```
int server_socket_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY; /* "any address I can use" */
    /* or: addr.s_addr.in_addr = INADDR_LOOPBACK (127.0.0.1) */
    /* or: addr.s_addr.in_addr = htonl(...); */
addr.sin_port = htons(9999); /* port number 9999 */

if (bind(server_socket_fd, &addr, sizeof(addr)) < 0) {
    /* handle error */
}
listen(server_socket_fd, 10); /* choose the number of unaccepted connections
...
int socket_fd = accept(server_socket_fd, NULL);
```

connection setup: client — manual addresses

```
int sock_fd;

server = /* code on later slide */;
sock_fd = socket(
    AF_INET, /* IPv4 */
    SOCK_STREAM, /* byte-oriented */
    IPPROTO_TCP
);
if (sock_fd < 0) { /* handle error */ }

struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = htonl(2156872459); /* 128.143.67.11 */
addr.sin_port = htons(80); /* port 80 */
if (connect(sock_fd, (struct sockaddr*) &addr, sizeof(addr)) {
    /* handle error */
}
DoClientStuff(sock_fd); /* read and write from sock_fd */
```

connection setup: client — manual addresses

```
int sock_fd;

server = /* code on later slide */;
sock_fd = socket(
    AF_INET, /* IPv4 */
    SOCK_STREAM, /* byte-oriented */
    IPPROTO_TCP
);
if (sock_fd < 0) { /* handle error */ }

/* specify IPv4 instead of IPv6 or local-only sockets
   specify TCP (byte-oriented) instead of UDP ('datagram' oriented) */
addr.sin_addr.s_addr = htonl(2156872459); /* 128.143.67.11 */
addr.sin_port = htons(80); /* port 80 */
if (connect(sock_fd, (struct sockaddr*) &addr, sizeof(addr)) {
    /* handle error */
}
DoClientStuff(sock_fd); /* read and write from sock_fd */
```


connection setup: client — manual addresses

```
int sock_fd;

server = /* code */
sock_fd = socket(AF_INET, /* SOCK_STREAM, /* byte-oriented */
                 IPPROTO_TCP
);
if (sock_fd < 0) { /* handle error */ }

struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = htonl(2156872459); /* 128.143.67.11 */
addr.sin_port = htons(80); /* port 80 */
if (connect(sock_fd, (struct sockaddr*) &addr, sizeof(addr)) {
    /* handle error */
}
DoClientStuff(sock_fd); /* read and write from sock_fd */
```

htonl/s = host-to-network long/short
network byte order = big endian

connection setup: client — manual addresses

```
int sock_fd;
```

```
server = / struct representing IPv4 address + port number  
sock_fd = declared in <netinet/in.h>  
          AF_INET see man 7 ip on Linux for docs  
          SOCK_STREAM  
          IPPROTO_TCP  
);  
if (sock_fd < 0) { /* handle error */ }
```

```
struct sockaddr_in addr;  
addr.sin_family = AF_INET;  
addr.sin_addr.s_addr = htonl(2156872459); /* 128.143.67.11 */  
addr.sin_port = htons(80); /* port 80 */  
if (connect(sock_fd, (struct sockaddr*) &addr, sizeof(addr)) {  
    /* handle error */  
}  
DoClientStuff(sock_fd); /* read and write from sock_fd */
```

echo client/server

```
void client_for_connection(int socket_fd) {
    int n; char send_buf[MAX_SIZE]; char recv_buf[MAX_SIZE];
    while (prompt_for_input(send_buf, MAX_SIZE)) {
        n = write(socket_fd, send_buf, strlen(send_buf));
        if (n != strlen(send_buf)) {...error?...}
        n = read(socket_fd, recv_buf, MAX_SIZE);
        if (n <= 0) return; // error or EOF
        write(STDOUT_FILENO, recv_buf, n);
    }
}



---


void server_for_connection(int socket_fd) {
    int read_count, write_count; char request_buf[MAX_SIZE];
    while (1) {
        read_count = read(socket_fd, request_buf, MAX_SIZE);
        if (read_count <= 0) return; // error or EOF
        write_count = write(socket_fd, request_buf, read_count);
        if (read_count != write_count) {...error?...}
    }
}
```

echo client/server

```
void client_for_connection(int socket_fd) {
    int n; char send_buf[MAX_SIZE]; char recv_buf[MAX_SIZE];
    while (prompt_for_input(send_buf, MAX_SIZE)) {
        n = write(socket_fd, send_buf, strlen(send_buf));
        if (n != strlen(send_buf)) {...error?...}
        n = read(socket_fd, recv_buf, MAX_SIZE);
        if (n <= 0) return; // error or EOF
        write(STDOUT_FILENO, recv_buf, n);
    }
}



---


void server_for_connection(int socket_fd) {
    int read_count, write_count; char request_buf[MAX_SIZE];
    while (1) {
        read_count = read(socket_fd, request_buf, MAX_SIZE);
        if (read_count <= 0) return; // error or EOF
        write_count = write(socket_fd, request_buf, read_count);
        if (read_count != write_count) {...error?...}
    }
}
```

echo client/server

```
void client_for_connection(int socket_fd) {
    int n; char send_buf[MAX_SIZE]; char recv_buf[MAX_SIZE];
    while (prompt_for_input(send_buf, MAX_SIZE)) {
        n = write(socket_fd, send_buf, strlen(send_buf));
        if (n != strlen(send_buf)) {...error?...}
        n = read(socket_fd, recv_buf, MAX_SIZE);
        if (n <= 0) return; // error or EOF
        write(STDOUT_FILENO, recv_buf, n);
    }
}



---


void server_for_connection(int socket_fd) {
    int read_count, write_count; char request_buf[MAX_SIZE];
    while (1) {
        read_count = read(socket_fd, request_buf, MAX_SIZE);
        if (read_count <= 0) return; // error or EOF
        write_count = write(socket_fd, request_buf, read_count);
        if (read_count != write_count) {...error?...}
    }
}
```

connection setup: server, address setup

```
/* example (hostname, portname) = ("127.0.0.1", "443") */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;

memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_INET; /* for IPv4 */
/* or: */ hints.ai_family = AF_INET6; /* for IPv6 */
/* or: */ hints.ai_family = AF_UNSPEC; /* I don't care */
hints.ai_flags = AI_PASSIVE;

rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }
```

connection setup: server, address setup

```
/* example (hostname, portname) = ("127.0.0.1", "443") */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;

memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_INET; /* for IPv4 */
/* or: */ hints.ai_family = AF_INET6; /* for IPv6 */
/* or: */ hints.ai_family = AF_UNSPEC; /* I don't care */
hints.ai_flags = AI_PASSIVE; /* hostname could also be NULL
                               means "use all possible addresses"
                               only makes sense for servers */

rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) {
```

connection setup: server, address setup

```
/* example (hostname, portname) = ("127.0.0.1", "443") */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;

memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_INET; /* for IPv4 */
/* or: */ hints.ai_family = AF_INET6; /* for IPv6 */
/* or: */ hints.ai_family = AF_UNSPEC; /* I don't care */
hints.ai_flags = 0;

rv = getaddrinfo(hostname, portname, &hints, NULL);
if (rv != 0) {
```

portname could also be NULL
means "choose a port number for me"
only makes sense for servers

connection setup: server, address setup

```
/* example (hostname, portname) = ("127.0.0.1", "443") */
const char *hostname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;

memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_INET; /* for IPv4 */
/* or: */ hints.ai_family = AF_INET6; /* for IPv6 */
/* or: */ hints.ai_family = AF_UNSPEC; /* I don't care */
hints.ai_flags = AI_PASSIVE;

rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }
```

AI_PASSIVE: "I'm going to use bind"

connection setup: server, addrinfo

```
struct addrinfo *server;  
... getaddrinfo(...) ...
```

```
int server_socket_fd = socket(  
    server->ai_family,  
    server->ai_socktype,  
    server->ai_protocol  
);
```

```
if (bind(server_socket_fd, ai->ai_addr, ai->ai_addr_len)) < 0) {  
    /* handle error */  
}  
listen(server_socket_fd, MAX_NUM_WAITING);  
...  
int socket_fd = accept(server_socket_fd, NULL);
```

connection setup: client, using addrinfo

```
int sock_fd;
struct addrinfo *server = /* code on next slide */;

sock_fd = socket(
    server->ai_family,
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...
    server->ai_socktype,
    // ai_socktype = SOCK_STREAM (bytes) or ...
    server->ai_protocol,
    // ai_protocol = IPPROTO_TCP or ...
);
if (sock_fd < 0) { /* handle error */ }
if (connect(sock_fd, server->ai_addr, server->ai_addrlen) < 0) {
    /* handle error */
}
freeaddrinfo(server);
DoClientStuff(sock_fd); /* read and write from sock_fd */
close(sock_fd);
```

connection setup: client, using addrinfo

```
int sock_fd;  
struct addrinfo *server = /* code on next slide */;  
  
sock_fd = socket(  
    server->ai_family,  
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...  
    server->ai_socktype,  
    // ai_socktype = SOCK_STREAM (bytes) or ...  
    server->ai_protocol,  
    //  
);  
if (sock_fd < 0) {  
    if (cor /* handles IPv4 and IPv6  
    /* handles DNS names, service names  
}  
freeaddrinfo(server);  
DoClientStuff(sock_fd); /* read and write from sock_fd */  
close(sock_fd);
```

connection setup: client, using addrinfo

```
int sock_fd;  
struct addrinfo *server = /* code on next slide */;  
  
sock_fd = socket(  
    server->ai_family,  
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...  
    server->ai_socktype,  
    // ai_socktype = SOCK_STREAM (bytes) or ...  
    server->ai_protocol,  
    // ai_protocol = IPPROTO_TCP or ...  
);  
if (sock_fd < 0) { /* handle error */ }  
if (connect(sock_fd, server->ai_addr, server->ai_addrlen) < 0) {  
    /* handle error */  
}  
freeaddrinfo(server);  
DoClientStuff(sock_fd); /* read and write from sock_fd */  
close(sock_fd);
```

connection setup: client, using addrinfo

```
int sock_fd;  
struct addrinfo *server_addr;  
  
sock_fd = socket(server->ai_family, server->ai_socktype,  
server->ai_protocol);  
// ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...  
// ai_socktype = SOCK_STREAM (bytes) or ...  
// ai_protocol = IPPROTO_TCP or ...  
);  
if (sock_fd < 0) { /* handle error */ }  
if (connect(sock_fd, server->ai_addr, server->ai_addrlen) < 0) {  
    /* handle error */  
}  
freeaddrinfo(server);  
DoClientStuff(sock_fd); /* read and write from sock_fd */  
close(sock_fd);
```

ai_addr points to struct representing address
type of struct depends whether IPv6 or IPv4

connection setup: client, using addrinfo

```
int sock_fd;
```

```
st
```

```
so
```

since addrinfo contains pointers to dynamically allocated memory,
call this function to free everything

```
    // ai_family = AF_INET (IPv4) or AF_INET6 (IPv6) or ...
    server->ai_socktype,
    // ai_socktype = SOCK_STREAM (bytes) or ...
    server->ai_protocol
    // ai_protocol = IPPROTO_TCP or ...
);
if (sock_fd < 0) { /* handle error */ }
if (connect(sock_fd, server->ai_addr, server->ai_addrlen) < 0) {
    /* handle error */
}
freeaddrinfo(server);
DoClientStuff(sock_fd); /* read and write from sock_fd */
close(sock_fd);
```

connection setup: lookup address

```
/* example hostname, portname = "www.cs.virginia.edu", "443" */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;
memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_UNSPEC; /* for IPv4 OR IPv6 */
// hints.ai_family = AF_INET4; /* for IPv4 only */

hints.ai_socktype = SOCK_STREAM; /* byte-oriented --- TCP */
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

/* eventually freeaddrinfo(result) */
```


connection setup: lookup address

```
/* example hostname, portname = "www.cs.virginia.edu", "443" */
const char *hostname; const char *portname;
...
struct addrinfo *server;
struct addrinfo hints;
int rv;
memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_UNSPEC; /* for IPv4 OR IPv6 */
// hints.ai_socktype = AF_INET; /* for TCP */
NB: pass pointer to pointer to addrinfo to fill in
hints.ai_socktype = SOCK_STREAM; /* byte-oriented --- TCP */
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

/* eventually freeaddrinfo(result) */
```

connection setup: lookup address

```
/* example hostname, portname = "www.cs.virginia.edu", "443" */
const
...
struct
struct
int rv;
memset(&hints, 0, sizeof(hints));
hints.ai_family = AF_UNSPEC; /* for IPv4 OR IPv6 */
// hints.ai_family = AF_INET4; /* for IPv4 only */

hints.ai_socktype = SOCK_STREAM; /* byte-oriented --- TCP */
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

/* eventually freeaddrinfo(result) */
```

connection setup: multiple server addresses

```
struct addrinfo *server;
...
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

for (struct addrinfo *current = server; current != NULL;
     current = current->ai_next) {
    sock_fd = socket(current->ai_family, current->ai_socktype, current->ai_protocol);
    if (sock_fd < 0) continue;
    if (connect(sock_fd, current->ai_addr, current->ai_addrlen) == 0)
        break;
}
close(sock_fd); // connect failed
}
freeaddrinfo(server);
DoClientStuff(sock_fd);
close(sock_fd);
```

connection setup: multiple server addresses

```
struct addrinfo *server;
...
rv = getaddrinfo(hostname, portname, &hints, &server);
if (rv != 0) { /* handle error */ }

for (struct addrinfo *current = server; current != NULL;
     current = current->ai_next) {
    sock_fd = socket(current->ai_family, current->ai_socktype, current->ai_protocol);
    if (sock_fd < 0) continue;
    if (connect(sock_fd, current->ai_addr, current->ai_addrlen) == 0)
        break;
}
close(sock_fd);
}
freeaddrinfo(server);
DoClient(sock_fd);
close(sock_fd);
```

addrinfo is a linked list

name can correspond to multiple addresses

example: redundant copies of web server

example: an IPv4 address and IPv6 address

connection setup: old lookup function

```
/* example hostname, portnum= "www.cs.virginia.edu", 443*/
const char *hostname; int portnum;
...
struct hostent *server_ip;
server_ip = gethostbyname(hostname);

if (server_ip == NULL) { /* handle error */ }

struct sockaddr_in addr;
addr.s_addr = *(struct in_addr*) server_ip->h_addr_list[0];
addr.sin_port = htons(portnum);
sock_fd = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
connect(sock_fd, &addr, sizeof(addr));
...
```

aside: on server port numbers

Unix convention: must be root to use ports 0–1023

root = superuser = 'administrator user' = what sudo does

so, for testing: probably ports > 1023